

Juwon Lee (Henna)

AI Product Engineer

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SUMMARY

A product engineer who takes ML-backed products from dataset to deployed interface, at the Seoul Robot & AI Science Museum. I build end to end, sensor to shipped interface, and keep it running where real users are. My Art Lab ran on the floor through June–July 2026, growing from 3,049 images to 8,000+; for Doctor-Green I built the training set myself — 39,889 lesion crops with a group-aware stratified split.

SKILLS

AI & Data	Python · TensorFlow.js · MobileNet v2 · VGG16 / CNN · OpenAI API · VQA Datasets · Dataset QA · AI Hub
Frontend & App	Next.js · React · TypeScript · JavaScript · Tailwind · PWA · Capacitor
Backend, Infra & Ops	Supabase · Firebase / Firestore · Flask · Serverless Functions · Cloudflare R2 · Vercel · Tailscale / WireGuard
Hardware · IoT · Robotics	ESP32 · MicroPython · Arduino / C++ · Sensors · 3D Printing
Planning & Education	Product Planning (PM) · Learning Design · Bilingual Docent (KO-EN)

EXPERIENCE

Education R&D · Developer 2026.03 – Present
Seoul Robot & AI Science Museum · Seoul, KR

- Research, build, and run AI education web apps live on the floor — the Art Lab ran there through June–July 2026, growing from 3,049 images to over 8,000.
- Design education programs end to end, including a smart-farm curriculum commissioned by Yangpyeong Education Office (full server & web-app stack) demo-taught to 5 high-school students in August 2026, and guide exhibitions bilingually (Korean / English).

VQA Dataset Builder & QA Reviewer 2025.11 – 2026.01
Select Star · Seoul, KR

- Built and reviewed VQA datasets for LLM training on an ETRI project, including complex 3-hop reasoning Q&A sets.
- Sustained about 2.5× the team's average throughput while holding accuracy.

Youth Intern 2025.04 – 2025.08
Ministry of Science and ICT · Sejong, KR

- Redesigned the national Pay-TV survey — what to measure and how to sample it, the same problem as dataset design; covered 11 ministry events.

SELECTED PROJECTS

Raimi's AI Art Lab — Museum service run Jun–Jul 2026: 3,049 images in June → 8,000+ in July, and no more unexplained freezes.

Tech JavaScript · OpenAI API · Vercel Serverless · Cloudflare R2 · Supabase

Doctor-Green — Live IoT monitoring and control paired with a crop-disease model rebuilt from the data up — 39,889 lesion crops built solo with a group-aware stratified split.

Tech Next.js 16 · TypeScript · Supabase · ESP32 · Python

Exhibit Auto-Recovery — An exhibit only one person could restart, now recoverable in one tap from any staff phone. 20s watchdog + mesh VPN.

Tech PowerShell · Windows Task Scheduler · HTTP service · Tailscale / WireGuard

AWARDS

- Prize · GH Youth Build-Up Start-up Competition — medication-management service “YAKMOA” 2026.03
- 2nd Place · Science Museum & Community AI Hackathon — AI docent 2025.09
- Excellence Award · ICT-SW Women's Start-up Competition — VGG16 diagnosis app 2024.12

LEADERSHIP & ACTIVITIES

- KB LA School (Middle School) — Middle-school Mentor, 2×/week 2026.07 – Present
- Gwangyang AI Smart i-Kium — Instructor, Grade-3 Math & English, 1×/week 2026.03 – 2026.08
- Geuruteogi Learning Mentoring “Gachi-Edu” — University Mentor, 2×/week 2026.02 – Present

EDUCATION

Hankyong National University 2022.03 – 2027.02 (expected)
B.Eng. in Software Convergence · B.Ag. in Plant Life & Environment (double major) · completing the degree while working full-time

Rutgers University — Intensive Winter English Program 2025.01 – 2025.02

Academic English & cross-cultural communication · New Brunswick, NJ, USA

Languages Korean (native) · English (fluent — TOEIC Speaking AL, OPIc AL)